學生教材

How Does a Game Place Sound Around You?

國中 7 年級　第 3 週　預計 94 分鐘

這週要帶走的能力

- 讀懂遊戲如何用位置、距離與遮蔽物改變玩家聽到的聲音。
- 學會 10 個 spatial audio 核心字。
- 用 at/on/in 與位置介系詞描述遊戲場景。
- 整合三段證據，判斷到底是哪一個空間關係改變。

第一遍只抓三個問題：聲音在哪裡？玩家在哪裡？中間有沒有東西？第二遍標出 distance、direction、obstruction 的證據。推論題先寫『改變的是哪一個關係』再選答案。

1 先自己讀　2 圈起生字　3 獨立作答　4 訂正思考　5 AI輔助解惑

先想一想

戴耳機玩遊戲時，如果怪物在你背後，你沒有回頭卻覺得聲音從後方來。你猜遊戲至少需要知道哪些位置資訊？

閱讀：How Does a Game Place Sound Around You?

情境背景：你在一個 3D 遊戲場景裡，前方有鈴鐺、右後方有怪物，中間還可能隔著牆。遊戲要怎麼讓這些聲音聽起來像從不同位置傳來？

讀法提示：每段旁邊先標 D：distance、Dir：direction、O：obstruction。；看到 farther/closer、behind/in front of、between 時，畫出兩個位置的關係。；題目如果同時改變兩個條件，不要急著說是哪一個造成結果。；產品或系統名稱只用來支撐文章中的精確功能，不需要背品牌。

In many games, sound is part of the map. **Spatial** audio places a sound **source** at a **position** in a 3D scene and lets a **listener** hear it from that location. The **source** might be a bell, a door, or a creature. The **listener** is the point from which the player hears the scene. An audio system can compare the **source** **position** with the **listener** **position** and change what the player hears. This makes location information useful even when the player is not looking at the **source**.

Distance changes the result. Apple's PHASE documentation explains that a sound farther from the **listener** can play more quietly, while a closer **source** can play more loudly. Roblox also describes positional audio whose volume changes with the **distance** between the **listener** and the sound **position**. This does not mean **distance** is the only setting in every game. It means **distance** can be one part of the audio system's decision.

Direction matters too. A **source** can be behind the **listener**, to the left, or in front. Apple notes that **spatial** audio can send more sound through the left speaker when a horn is on the left side of a scene. Meta's **Spatial** SDK says a sound can stay in its virtual **position** when the **listener** turns their head. So if a sound starts behind you, turning your view does not require the virtual **source** itself to move.

The **environment** can add another layer. Apple's PHASE can model **obstructions** and indoor sound **reflections**. Its overview gives an example in which the volume of an incoming fireball is lowered when the player takes cover behind a wall. The important point is not that every wall works the same way in every game. The system may use an **obstruction** or room shape as information when it changes the sound.

TEST SCENE

At 8:00 in the training room, stand at the center point. A bell is on a table in front of you. A creature is behind a wall to your right. First, keep the bell in the same **position** and walk nearer to it. Next, return to the center and turn around. Finally, move behind the wall without moving the creature. Record what changes in **distance**, **direction**, or **obstruction**.

A careful test changes one relationship at a time. Moving nearer changes **distance**. Turning around changes the **listener**'s **orientation** relative to the **source**. Stepping behind a wall introduces an **obstruction**. These changes can produce different audio results, so a good explanation should name the exact relationship that changed. **Spatial** audio does not need to understand danger like a human. It can react to scene data about where the **source** and **listener** are and what lies between them.

核心單字

spatial *(adj.)* 空間的；與位置有關的
Spatial audio uses location information.
空間音訊會使用位置資訊。

source *(n.)* 來源；聲音來源
The bell is the sound source.
鈴鐺是聲音來源。

listener *(n.)* 聽者；接收聲音的位置
The player is the listener in this scene.
玩家在這個場景中是聽者。

position *(n.)* 位置
The source stays in the same position.
聲音來源保持在同一個位置。

distance *(n.)* 距離
The distance becomes shorter when you walk nearer.
走近時距離會變短。

direction *(n.)* 方向
Direction tells you where a sound is around you.
方向告訴你聲音在你周圍的哪一邊。

orientation *(n.)* **朝向；面對的方向**

Turning around changes the listener's orientation.

轉身會改變聽者的朝向。

obstruction *(n.)* **阻擋物；遮蔽**

A wall can act as an obstruction between two points.

牆可以成為兩個位置之間的遮蔽物。

reflection *(n.)* **反射；反射回來的聲音**

A room can add sound reflections.

房間可能加入聲音反射。

environment *(n.)* **環境**

The environment can affect spatial sound.

環境可能影響空間聲音。

觀念解說：時間與位置介系詞：先判斷是點、面、裡面，還是相對位置

at 常用在明確時間點或地點點位，例如 at 8:00、at the gate；on 常用在某一天或物體表面，例如 on Friday、on the table；in 常用在月份、年份或一個空間裡，例如 in September、in the room。描述相對位置時可用 behind、in front of、near、to the left/right of。先問自己『我要描述哪一種關係』，再選介系詞。

> **核心句型結構**
>
> **at + exact time / point**

> **核心句型結構**
>
> **on + day / surface**

> **核心句型結構**
>
> **in + month / year / enclosed place**

> **核心句型結構**
>
> **behind / in front of / near / to the left of / to the right of**

完整示範

At 8:00, the player is in the training room.

解析思考：8:00 是精確時間點，用 at；training room 是人在其中的空間，用 in。

The bell is on the table in front of the player.

解析思考：bell 接觸桌面用 on；相對於玩家在前方，用 in front of。

容易踩到的盲點與陷阱

✗ 錯誤 ~~The test starts in 8:00.~~

✓ 正確 **The test starts at 8:00.**

💡 **為什麼：**精確鐘點用 at，不用 in。

✗ 錯誤 ~~The bell is in the table.~~

✓ 正確 **The bell is on the table.**

💡 **為什麼：**放在桌面上是 on；in the table 會變成在桌子裡面。

觀念解說：空間推論：先找『哪一個關係變了』

文章同時有距離、方向與遮蔽。作答時先固定 source，再問 listener 有沒有移動或轉向、兩者中間有沒有多一面牆。若同時改變 distance 和 obstruction，就不能只靠結果斷言是哪一個單獨造成變化。

> **核心句型結構**
>
> **same source + different distance**

> **核心句型結構**
>
> **same positions + different orientation**

> **核心句型結構**
>
> **same source + obstruction between source and listener**

完整示範

The bell stays in place, but the player walks nearer.

解析思考：source position 沒變，主要改變的是 distance，因此可以用距離段落找證據。

The player walks nearer and also moves behind a wall.

解析思考：distance 與 obstruction 同時變，不能只把音訊差異歸因給其中一個。

容易踩到的盲點與陷阱

✗ 錯誤 ~~The sound changed, so distance must be the reason.~~

✓ 正確 **Check whether distance, direction, or obstruction changed before naming the reason.**

💡 **為什麼：**結果改變不代表只改了一個條件；先檢查關係才能避免過度推論。

跟著示範 **引導練習：時間、位置、距離**

先判斷是在考介系詞、核心字，還是正文證據。閱讀題一定回原文找句子。

> **提示：**精確時間用 at；物體表面常用 on。

G1

The bell is ___ the table in the test scene.

(A) on

(B) at

(C) in

(D) behind

G2

In paragraph 1, what is the listener?

(A) The wall between two sounds

(B) The point from which the player hears the scene

(C) The loudest sound source

(D) The program that creates the game map

G3

The bell stays in the same position, but the player walks nearer to it. Which relationship clearly changes?

(A) The bell's instrument type

(B) The wall shape

(C) The distance between source and listener

(D) The time of the test

自己完成

獨立練習：三種空間關係整理

先完成 I1 organizer，再做推論題。每列都要從正文找一個 action 與一個 system response/effect。

I1

Complete the organizer for distance, direction, and obstruction. For each relationship, write one action that changes it and one reading-supported audio effect or system response.

Relationship	Action that changes it	Reading-supported effect / response
Distance		
Direction / orientation		
Obstruction		

I2

A creature stays in the same position while the player turns around. What changed most directly?

(A) The creature became closer

(B) The listener's orientation relative to the source

(C) A new wall appeared

(D) The sound source moved to another room

I3

In paragraph 4, obstruction most nearly means ____.

(A) a sound that becomes music

(B) a point used to measure time

(C) something between positions that can block or affect the sound

(D) a command that moves the listener

會考型轉移

會考素養轉移：不要把兩個改變說成一個原因

每題先寫出關鍵關係。主旨題要涵蓋 distance、direction、environment；比較題要確認兩段證據都成立。

> **提示：**看到 always、only、must 時，檢查文章是否真的支持這麼強的說法。

C1

What is the main idea of the reading?

(A) Every game uses exactly the same spatial-audio system.

(B) Spatial audio can use source-listener relationships such as distance, direction, and obstruction to change what a player hears.

(C) A player should always wear headphones to hear a game correctly.

(D) Only walls can change the volume of a game sound.

C2

What do the Apple and Roblox examples in paragraph 2 both support?

(A) Distance between listener and source can influence the volume of positional audio.

(B) Every distant sound becomes completely silent.

(C) Direction never matters when distance changes.

(D) All games use Apple's PHASE system.

C3

A player walks nearer to a creature and also moves behind a wall. The sound changes. Why is it too strong to say that distance alone caused the change?

(A) Because the creature must have moved too.

(B) Because direction can never change in a game.

(C) Because both distance and obstruction changed, so the test does not isolate one relationship.

(D) Because spatial audio ignores scene information.

寫出自己的答案

句子產出：把位置與原因說精確

句數是題目條件的一部分。先寫位置，再寫原因；不要加入正文沒有支持的功能。

提示：可用 At 8:00... / in the room / on the table / behind the wall / because...

P1

Write exactly two complete sentences describing the TEST SCENE. Use at least three place/time prepositions or position phrases from this week's lesson.

P2

Write exactly two complete sentences explaining one audio change with because. Name the relationship that changed and one condition that stayed the same.

隔天提取

隔天提取：新文法 + 舊命令句

先遮住前面內容完成，再回去核對。R2 是上週命令句的 spaced review，不是重新學一次。

R1

The test begins ___ 8:00 ___ the training room.

(A) in / at

(B) on / on

(C) at / in

(D) at / on

R2

Write exactly three imperative sentences for a fair distance test. Keep the source in the same position and change only the listener's distance.

自我檢核

完成今日學習後，請勾選已掌握的項目，並寫下心得或問題：

- ☐ 我能分辨 at/on/in 的時間與地點用法。
- ☐ 我能用 behind、in front of、near、to the left/right of 描述相對位置。
- ☐ 我能分開 distance、direction/orientation、obstruction 三種關係。
- ☐ 同時改變兩個條件時，我不會硬說只有一個原因。
- ☐ 我能把上週的命令句用在完整三步驟，而不是只選單句答案。

帶走一點，隔天再想一次

任務目標：分兩天完成。第一天做 H1、H2，第二天做 H3。先靠記憶作答，再回正文標出支持句。（預計 11 分鐘）

建議在完成教材隔天再作答，測試自己是否真正記住。

H1

The game event is ___ Friday ___ 7:30 p.m.

(A) at / on
(B) on / at
(C) in / at
(D) on / in

H2

A sound source moves from in front of the player to behind the player. The distance stays the same and there is no wall. Which relationship clearly changed?

(A) Distance only
(B) Obstruction only
(C) Time only
(D) Direction relative to the listener

H3

Write exactly three complete sentences explaining why a test that changes both distance and wall obstruction cannot prove which one alone caused an audio change. Use at least one specific idea from the reading.